

Wireless Bluetooth Upgrade of the FullSoul Pressure Sensing Insole

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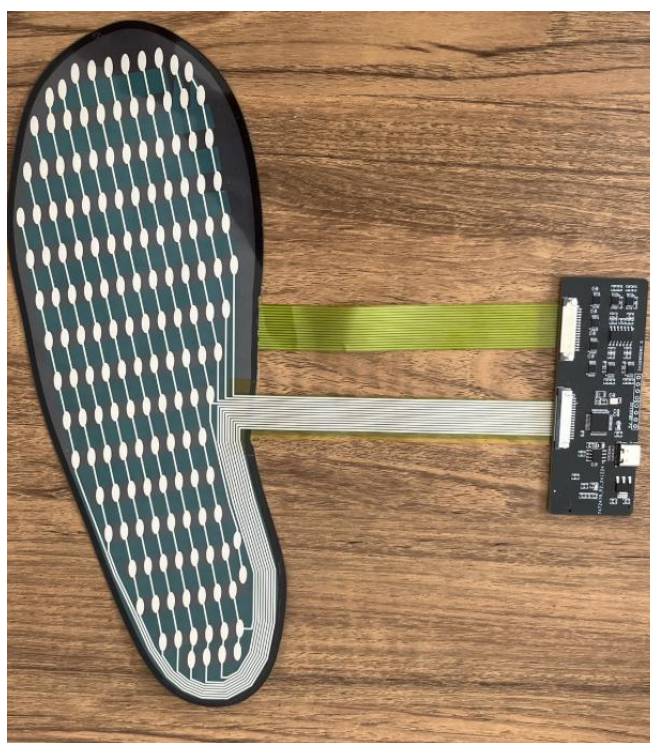
1. Motivation

- Animals naturally walk barefoot.
- Modern shoes reduce sensory feedback.
- Thin leather soles preserve natural foot sensation.
- Pressure measurements to evaluate this hypothesis.



2. Previous Work

- Used a proprietary microcontroller supplied with sensor
- Wired connection to a computer. [1]



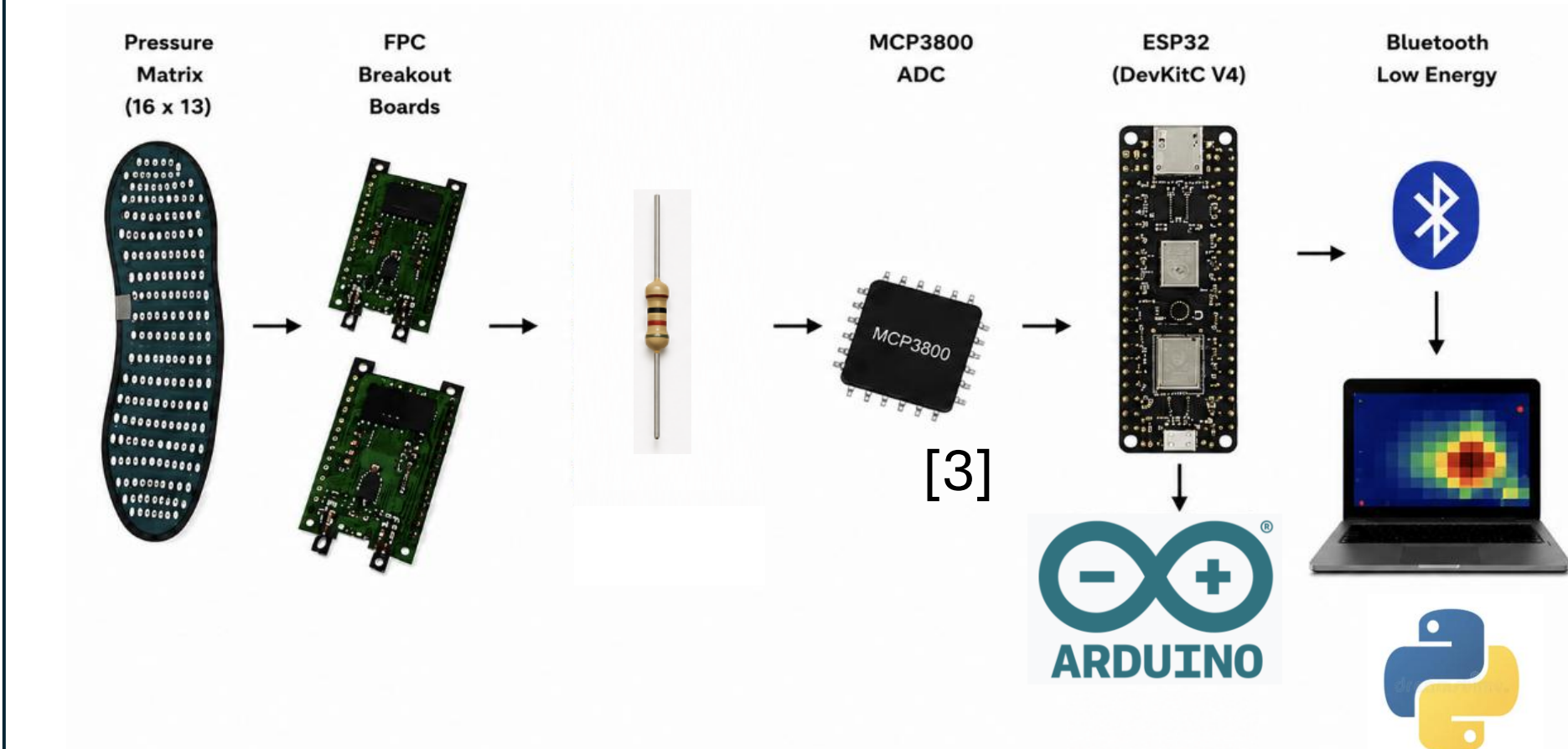
✗ Not portable, cannot be used wirelessly.

Next step: Adapt the system for Bluetooth communication.

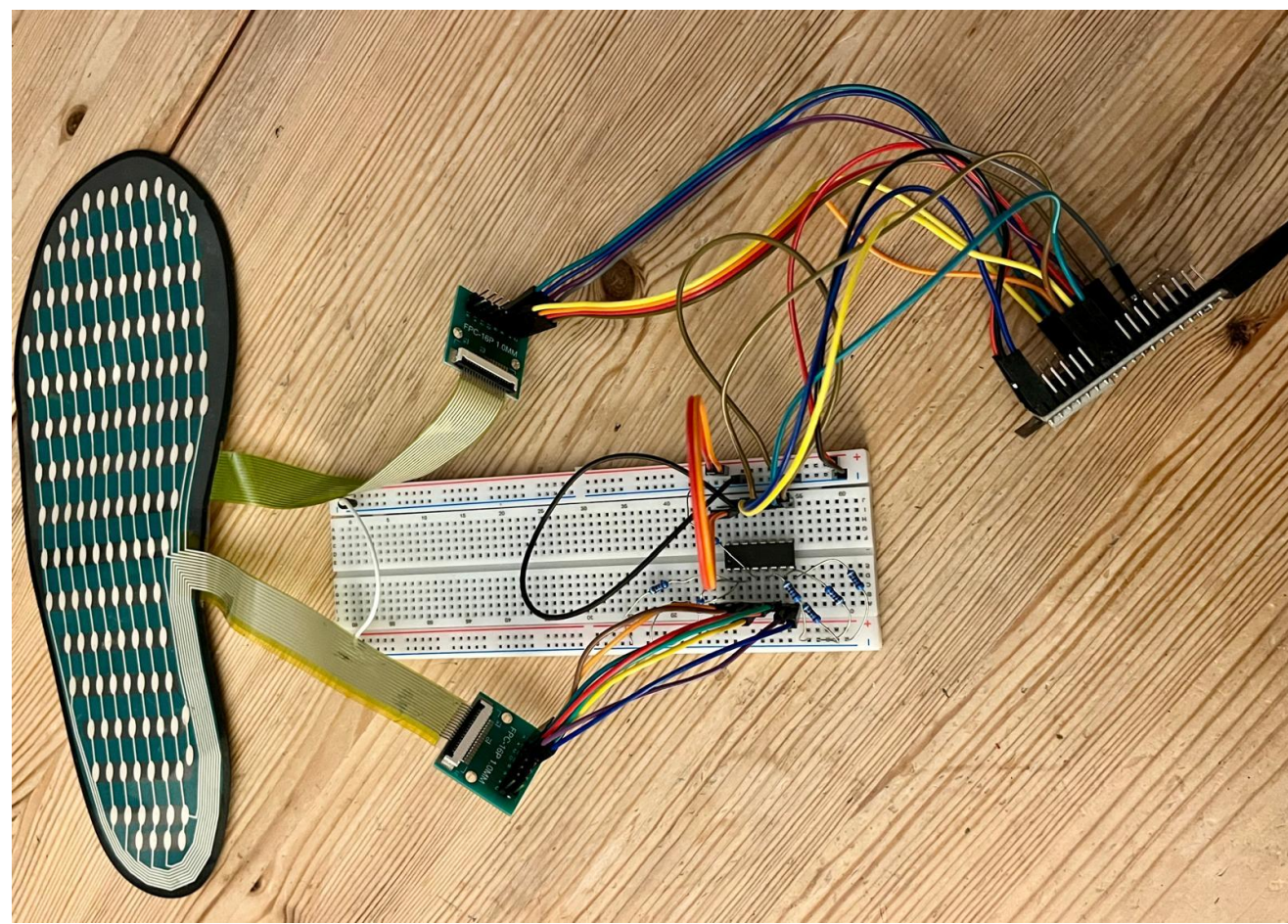
3. My Contribution

- ✓ Replaced controller (ESP32) [2]
- ✓ Bluetooth (BLE) communication
- ✓ Python real-time visualization
- ✓ Portable, open-source prototype

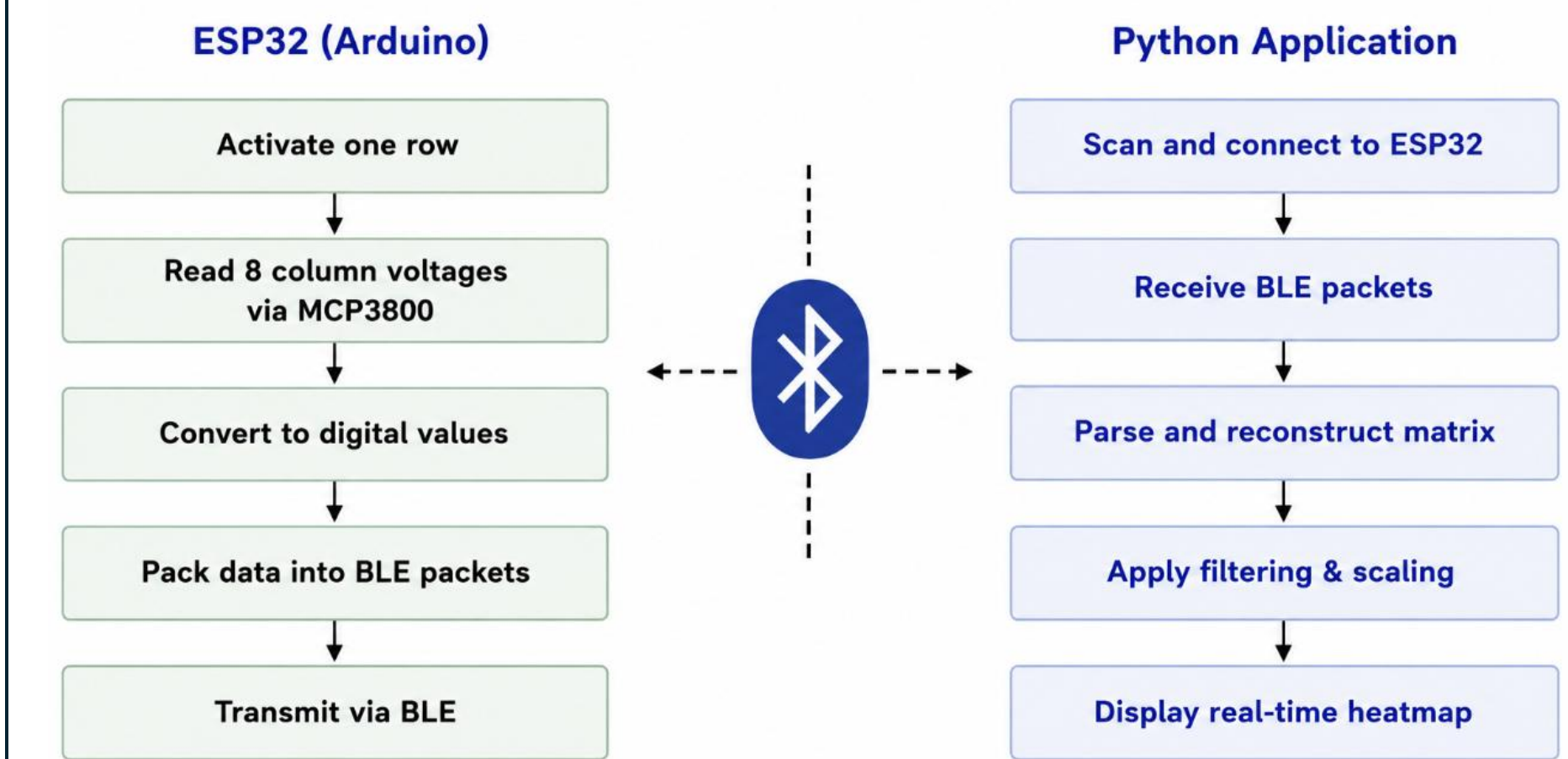
4. System Overview



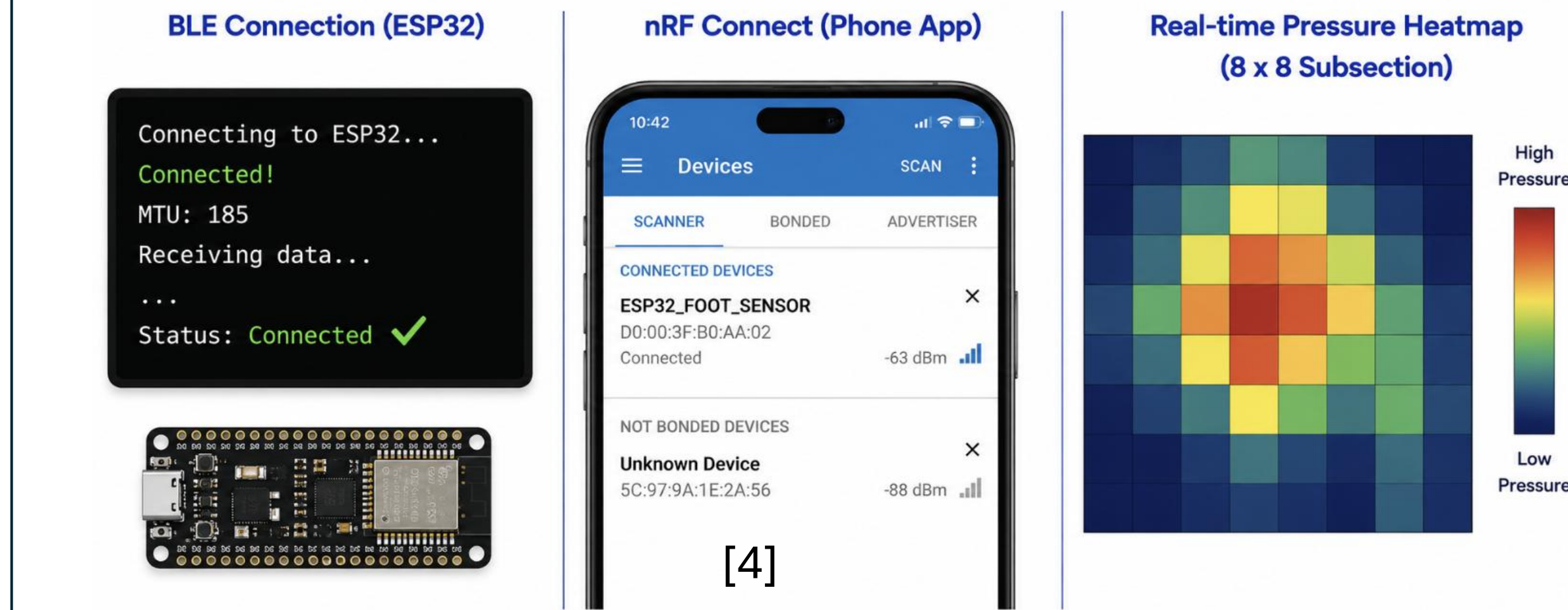
5. Hardware



6. Software



7. Results



8. Discussion

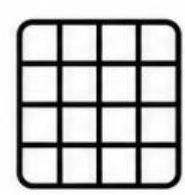
Advantages

- ✓ Wireless
- ✓ Low cost
- ✓ Portable

Limitations

- ✗ 8 x 8 prototype
- ✗ Calibration required
- ✗ Breadboard implementation

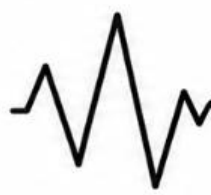
9. Future Work



Full 16 x 13
matrix
acquisition



Improve
calibration &
linearization



Increase
resolution &
sampling rate



Advanced
analytics &
gait analysis

10. References

- [1] Burian, K. FullSoul Running Pad...
- [2] ESP32 DevKitC V4 User Guide
- [3] MCP3008 Datasheet
- [4] nRF Connect Documentation

11. Acknowledgements

I thank Prof. Martin Daumer, the CACoM teaching team, and Kai Burian for providing the original FullSoul prototype that served as the foundation for this project.